

Algebra 1
Unit 4
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

A study of the average emissions of carbon dioxide (CO_2) from a vehicle in the Netherlands was conducted from 2000 to 2019. The function $C(x) = -5x + 190$ measures the carbon intensity of newly registered passenger vehicles in the Netherlands where x is the number of years since 2000 and C is the average emissions of CO_2 (in grams) per kilometer.

1. Two students were discussing how to find the domain of the function $C(x) = -5x + 190$. Their explanations are shown.

Michelle	Jaheim
The domain is all real numbers because the function is linear.	The domain should start at 0 because x represents the number of years since 2000. It would make sense for the domain to end with 19 because that is the last year of the study.

Determine which student is correct and write the domain using inequalities.

Jaheim is correct.

$$0 \leq x \leq 19$$

2. Write the range of the function $C(x) = -5x + 190$ using inequalities.

$$95 \leq y \leq 190$$

3. Determine the x -intercept.

$$-5x + 190 = 0$$

$$-5x = -190$$

$$x = 38$$

$$x\text{-intercept} = 38$$

4. Interpret the x -intercept.

In the year 2038, the measure of carbon intensity for newly registered passenger vehicles in the Netherlands will reach an average emission rate of 0 CO_2 per kilometer.

5. Determine the y -intercept.

$$\begin{aligned} c(0) &= -5(0) + 190 \\ &= 190 \end{aligned}$$

$$y\text{-intercept} = 190$$

6. Interpret the y -intercept.

At the beginning of the study, in the year 2000, the average carbon emission rate for newly registered passenger vehicles in the Netherlands was measured to be 190 CO₂ per kilometer.

7. Complete the statement about the line $C(x) = -5x + 190$.

As x -values increase, y -values decrease

8. Determine the slope.

$y = mx + b$, m represents the slope

$$m = -5$$

$$\text{slope} = -5$$

9. Interpret the slope.

The average carbon emission rate for newly registered passenger vehicles in the Netherlands decreases by 5 CO₂ per kilogram for each successive year after 2000.

10. Graph the function.

